

Multi-App Hosting Using AWS Network Load Balancer (NLB)

Introduction

Network Load Balancer (NLB) is a Layer 4 load balancing service provided by AWS that distributes incoming TCP, UDP, and TLS traffic across multiple targets such as EC2 instances. It is designed for high performance, low latency, and handling millions of requests per second.

This project demonstrates hosting multiple applications on different EC2 instances and distributing traffic using a Network Load Balancer.

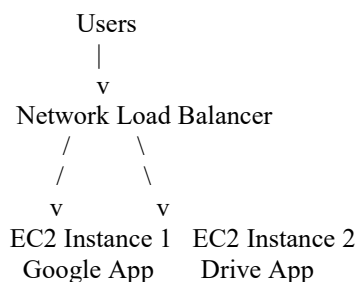
Objective

The objective of this project is to deploy multiple web applications on separate EC2 instances and distribute incoming traffic through a Network Load Balancer to ensure high availability, fault tolerance, and scalability.

AWS Services Used

- Amazon EC2
- Network Load Balancer (NLB)
- Target Groups
- Security Groups
- Amazon VPC
- GitHub (Application Source Code)

Architecture



Working Principle

1. Users access applications using the NLB DNS name.
2. NLB receives TCP traffic on port 80.
3. Traffic is forwarded to registered EC2 instances.
4. Health checks continuously monitor instance availability.
5. If one instance becomes unavailable, traffic is routed to healthy targets.
6. Applications remain accessible without interruption.

What is Network Load Balancer?

Network Load Balancer operates at Layer 4 of the OSI model and distributes TCP/UDP traffic across multiple targets.

Features

- High Performance
- Low Latency
- Static IP Support
- High Availability
- Automatic Failover
- Multi-AZ Support

Target Groups

Target Groups contain registered EC2 instances that receive traffic from the NLB.

Benefits

- Centralized traffic management
- Health monitoring
- Fault tolerance
- High availability

Health Checks

Health checks verify whether EC2 instances are available and capable of serving requests.

Health Check Configuration

Protocol : TCP

Port : 80

Healthy Target

TCP connection successful

Unhealthy Target

Port unavailable

Service stopped

Instance unavailable

High Availability

The project deploys applications across multiple Availability Zones.

Benefits:

- Fault Tolerance
- Reduced Downtime
- Increased Reliability
- Better User Experience

Difference Between ALB and NLB

Feature	ALB	NLB
Layer	7	4
Path-Based Routing	Yes	No
Host-Based Routing	Yes	No
TCP/UDP Support	Limited	Yes
Static IP	No	Yes
Performance	High	Very High

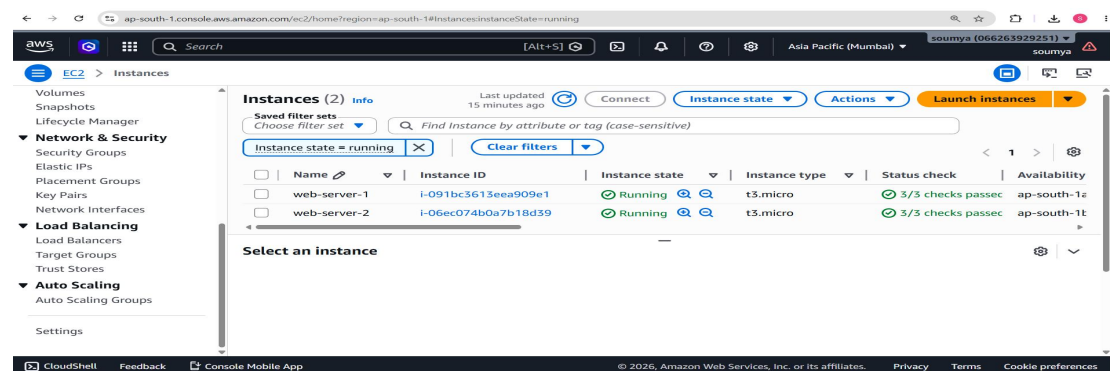
Project Outcome

- Hosted multiple applications on EC2 instances.
- Distributed traffic using Network Load Balancer.
- Implemented health checks.
- Achieved high availability.
- Demonstrated load balancing concepts.

Conclusion

AWS Network Load Balancer provides highly scalable and fault-tolerant traffic distribution for TCP/UDP applications. By combining EC2 instances, Target Groups, and Health Checks, organizations can build highly available and resilient infrastructures.

Screenshots:



ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:066263929251:targetgroup/tar-gp/b1b904e19246ed54

EC2 > Target groups > tar-gp

tar-gp

Details

arn:aws:elasticloadbalancing:ap-south-1:066263929251:targetgroup/tar-gp/b1b904e19246ed54

Target type Instance	Protocol : Port TCP: 80	VPC vpc-060381ef95bb4bf36	IP address type IPv4
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Load balancer
[Network-LoadB](#)

Total targets	Healthy	Unhealthy	Unused	Initial	Draining
2	1	1	0	0	0

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:ap-south-1:066263929251:loadbalancer/net/Network-LoadB/c6812861bf4633bd

EC2 > Load balancers > Network-LoadB

Network-LoadB

Details

Load balancer type Network	Status Active	VPC vpc-060381ef95bb4bf36	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZVDRBQ08TROA	Availability Zones subnet-02e317354667294ba ap-south-1b (aps1-az3) subnet-03f48ac0b11b6b52c ap-south-1a (aps1-az1)	Date created June 11, 2026, 17:46 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:066263929251:loadbalancer/net/Network-LoadB/c6812861bf4633bd	DNS name info Network-LoadB-c6812861bf4633bd.elb.ap-south-1.amazonaws.com (A Record)		

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